AMEEN SIDDIQUI

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Education

Master of Science in Process Engineering and Energy Technology, University of Applied Sciences, Bremerhaven

Experience

Data Scientist, Lead - KifferAI

02/2024 - 06/2025

Bengaluru

- Led a 3-person team to **Design** and **Deploy advanced machine learning models** for an automated e-commerce image processing pipeline, integrating Computer Vision and **Generative AI** (**LLM** concepts applied contextually).
- Engineered a sub-second background removal microservice using ONNX-optimized U2-Net, significantly reducing **Supply chain optimization** costs by cutting AWS EC2 inference costs by 75%.
- **Developed** and **Deploy advanced machine learning models** for zero-shot classification (CLIP/AWS Rekognition) to automate product categorization, eliminating manual data labeling efforts.
- **Build scalable solutions** by creating a **Generative AI** ad-creation engine using ComfyUI for contextual backgrounds, delivering a live Hugging Face POC for rapid stakeholder evaluation.
- **Developed** and containerized an asynchronous FastAPI backend on AWS (EC2/S3) to efficiently **Build scalable solutions** supporting high-concurrency mobile application traffic.

Machine Learning Researcher - Evonik Industries AG

05/2022 - 09/2023

Marl, Germany

- **Deployed advanced machine learning models** for a global **Demand forecasting** system covering 500 Make-to-Order polymers (12-month horizon), achieving a 20% reduction in WMAPE to optimize capacity planning and **Supply chain optimization**.
- Architected end-to-end MLOps pipelines, orchestrating models via SAP Data Intelligence and visualizing insights in SAP SAC, demonstrating ability to **Build scalable solutions** within enterprise ecosystems.
- Engineered a two-stage dimensionality reduction pipeline (LASSO + Tree-based importance) to distill 2,800 lagged macroeconomic variables into the top 10 predictors for **Forecasting** accuracy.
- Implemented Expanding Window Cross-Validation with custom indicator variables to prevent temporal data leakage and isolate macroeconomic shocks, enhancing **Time series analysis** robustness.
- **Developed** a point-wise **Time series analysis** anomaly detection model for extrusion lab data (pressure/temp) to ensure manufacturing quality and process stability.

Data Analyst - Sportsdata AG

01/2018 - 04/2022

North Germany

- Implemented an Insights Dashboard for Football event data, supporting real-time mobile application performance.
- **Developed** an Expected Goals model (xG) utilizing **Time series analysis** principles for offensive and defensive actions, improving prediction metrics by 15-20%.
- Analyzed event-data across more than 250 matches, providing foundational data for future **Forecasting** initiatives.
- Collaborated in a multicultural, cross-functional team environment, translating data insights for product managers and broadcasters.

Research Assistant - Technologie-Transfer-Zentrum (ttz) Bremerhaven

09/2014 - 12/2017

Bremerhaven, Germany

- Managed the National Bio-gas Project in East Germany, involving project planning and stakeholder reporting.
- Performed mathematical modelling and **Simulations** of the kinetics of the two-stage Biogas process, applying **Operations research** concepts.
- Conducted and maintained lab and pilot scale experiments, ensuring data integrity for subsequent analysis.
- Collected, cleaned, and transformed experimental data, preparing inputs for **Time series models**.

Senior Engineer - Kirloskar Oil Engines Ltd

08/2011 - 08/2013

Pune, India

- Led market expansion efforts in Southern India for new applications of existing product lines.
- Assessed technical requirements and engine parameters for new applications, ensuring product fit.
- Conducted engine delivery operations and third-party engine inspections for OEM clients.
- Coordinated production planning and delivery schedules with OEM partners and Line managers.

Skills

Machine_Learning_and_Modeling, Technologies_and_Tools, Optimization_and_Operations

Achievements

- Reduced AWS EC2 inference costs by 75% by engineering a sub-second background removal microservice, demonstrating strong **Operations research** application in cloud cost management.
- Eliminated the need for manual data labeling by **Deploying advanced machine learning models** using a zero-shot classification system (**Generative AI** integration).
- Reduced WMAPE by 20% in global **Demand forecasting** system to optimize capacity planning, directly impacting **Supply chain optimization**.
- Improved Expected Goals model (xG) prediction metrics by 15-20% using advanced **Time series analysis** techniques.